

Advanced Machine Learning

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Lab 4

1. The approaches to measure performance differ significantly between classification and clustering
2. Hierarchical Clustering



The approaches to measure performance in classification and clustering

Classification:

- **Objective:** Assign each instance to a predefined class based on a labeled dataset.
- **Metrics:** Since labels are known, accuracy and similarity to true labels are central to evaluating performance.

Clustering:

- **Objective:** Group similar instances without predefined labels, aiming to find natural clusters within the data.
- **Metrics:** As labels are unknown, evaluation focuses on intra-cluster cohesion and inter-cluster separation, as well as some indices if ground truth labels are available.



The approaches to measure performance in classification and clustering

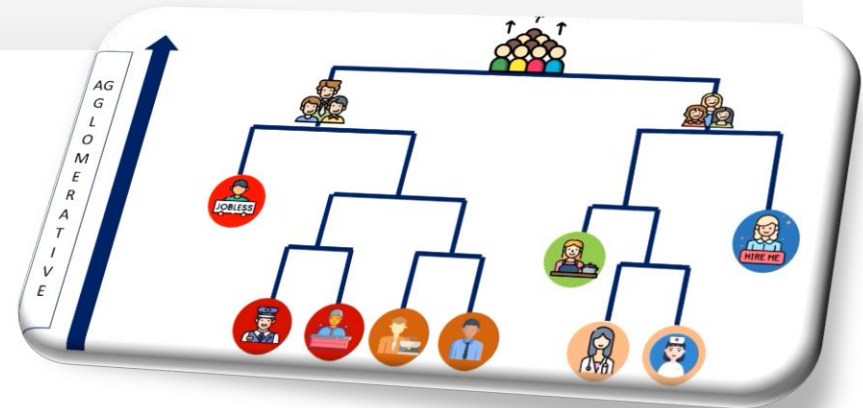
Task Type	Metric	Description
Classification	Accuracy, Precision, Recall, F1	Measures correct predictions vs. true labels.
Classification	Confusion Matrix	Shows misclassifications across classes.
Classification	ROC-AUC, Log-Loss	Measures model confidence and probability.
Clustering	Silhouette Score, Inertia, Dunn Index	Measures compactness and separation of clusters.
Clustering	Adjusted Rand Index, NMI, FMI	External metrics if true labels are available.

In summary, **classification** metrics rely on comparing predictions to true labels, while **clustering** metrics focus on the data's structure and grouping quality, often without ground truth.



What is Hierarchical Clustering

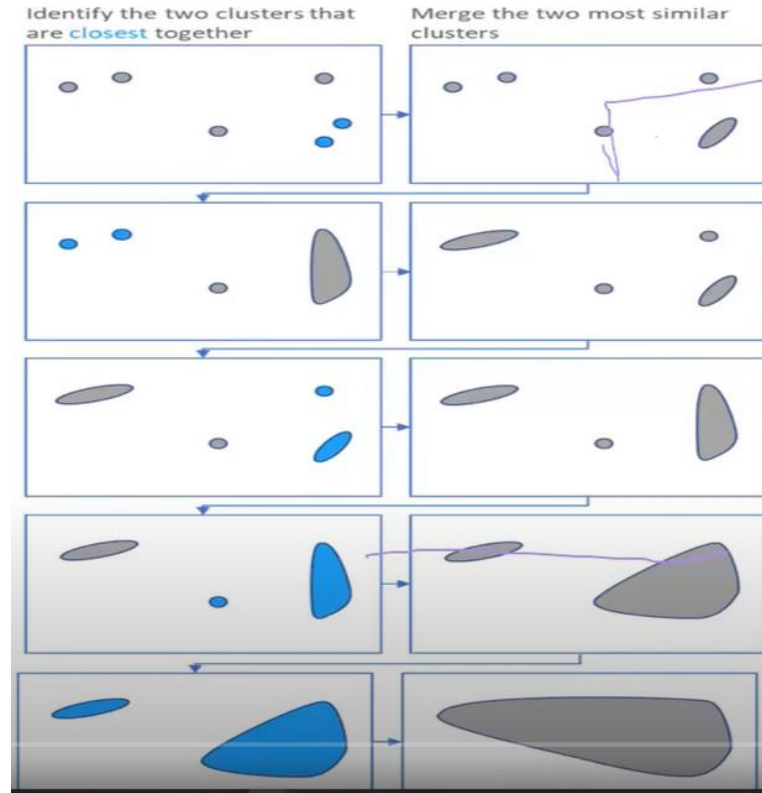
- Is an algorithm that groups similar objects into groups called *clusters*.
- The endpoint is a set of clusters, where each cluster is distinct from each other cluster, and the objects within each cluster are broadly similar to each other.



How it works

- ❑ Hierarchical clustering starts by treating each observation as a separate cluster.
- ❑ Then, it repeatedly executes the following two steps:
 - Identify the two clusters that are closest together.
 - Merge the two most similar clusters.
 - This iterative process continues until all the clusters are merged together.

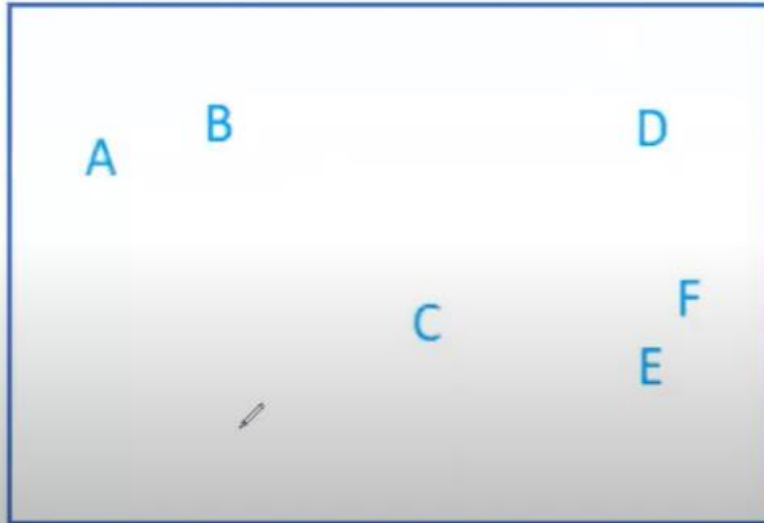
Hierarchical Clustering



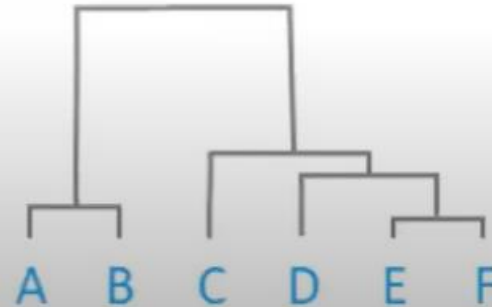
Hierarchical Clustering

The output

The main output of Hierarchical Clustering is a dendrogram, which shows the hierarchical relationship between the clusters:



Dendrogram



Hierarchical Clustering

Measures of distance (similarity)

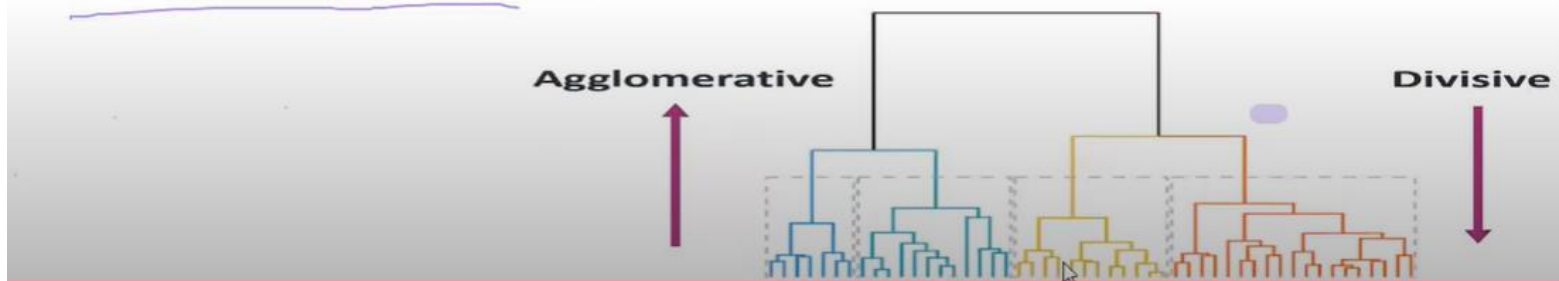
- ❑ In the previous example, the *distance* between two clusters has been computed based on the length of the straight line drawn from one cluster to another.
- ❑ This is commonly referred to as the *Euclidean distance*.
- ❑ Many other *distance metrics* have been developed.



Hierarchical Clustering

Agglomerative Vs. divisive algorithms

- ❑ Hierarchical clustering typically works by sequentially merging similar clusters, as shown above. This is known as *agglomerative hierarchical clustering*.
- ❑ In theory, it can also be done by initially grouping all the observations into one cluster, and then successively splitting these clusters.
- ❑ This is known as *divisive hierarchical clustering*.
- ❑ Divisive clustering is rarely done in practice.



Hierarchical Clustering Example

Read
Mall_Customers
dataset

```
CustomerID,Genre,Age,Annual Income (k$),Spending Score (1-100)
0001,Male,19,15,39
0002,Male,21,15,81
0003,Female,20,16,6
0004,Female,23,16,77
0005,Female,31,17,40
0006,Female,22,17,76
0007,Female,35,18,6
0008,Female,23,18,94
0009,Male,64,19,3
0010,Female,30,19,72
0011,Male,67,19,14
0012,Female,35,19,99
0013,Female,58,20,15
0014,Female,24,20,77
0015,Male,37,20,13
0016,Male,22,20,79
0017,Female,35,21,35
0018,Male,20,21,66
0019,Male,52,23,29
0020,Female,35,23,98
0021,Male,35,24,35
0022,Male,25,24,73
0023,Female,46,25,5
0024,Male,31,25,73
0025,Female,54,28,14
0026,Male,29,28,82
0027,Female,45,28,32
0028,Male,35,28,61
0029,Female,40,29,31
0030,Female,23,29,87
0031,Male,60,30,4
0032,Female,21,30,73
0033,Male,53,33,4
0034,Male,18,33,92
0035,Female,49,33,14
0036,Female,21,33,81
```



Hierarchical Clustering Cont.

Importing the libraries

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

[1] ✓ 6.4s

Importing the dataset

```
dataset = pd.read_csv('Mall_Customers.csv')
X = dataset.iloc[:, [3, 4]].values
```

✓ 0.0s

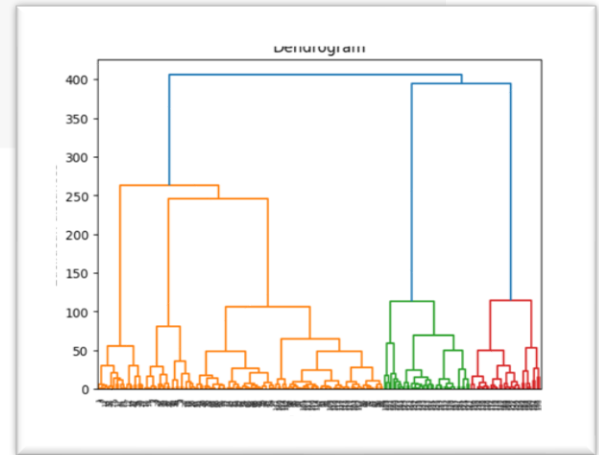
```
X
✓ 0.0s
array([[ 15,  39],
       [ 15,  81],
       [ 16,   6],
       [ 16,  77],
       [ 17,  40],
       [ 17,  76],
       [ 18,   6],
       [ 18,  94],
       [ 19,   3],
       [ 19,  72],
       [ 19,  14],
       [ 19,  99],
       [ 20,  15],
       [ 20,  77],
       [ 20,  13],
       [ 20,  79],
       [ 21,  35],
       [ 21,  66],
       [ 23,  29],
       [ 23,  98],
       [ 24,  35],
       [ 24,  73],
       [ 25,   5],
       [ 25,  73],
       [ 28,  14],
       ..
       [120,  79],
       [126,  28],
       [126,  74],
       [137,  18],
       [137,  83]], dtype=int64)
```



Hierarchical Clustering Cont.

- ✓ Using the dendrogram to find the optimal number of clusters

```
[3] import scipy.cluster.hierarchy as sch
dendrogram = sch.dendrogram(sch.linkage(X, method = 'ward'))
plt.title('Dendrogram')
plt.xlabel('Customers')
plt.ylabel('Euclidean distances')
plt.show()
```



Hierarchical Clustering Cont.

- **`sch.linkage(X, method='ward')`**: Creates a hierarchical cluster tree using the Ward method, which minimizes the sum of squared differences within clusters.
- **`sch.dendrogram(...)`**: Generates the dendrogram, which helps to visualize how clusters are merged at each step.
- **`plt.show()`**: Displays the dendrogram plot.



Hierarchical Clustering Cont.

- ✓ Training the Hierarchical Clustering model on the dataset

```
[5] from sklearn.cluster import AgglomerativeClustering  
    hc = AgglomerativeClustering(n_clusters = 5, metric = 'euclidean', linkage = 'ward')  
    y_hc = hc.fit_predict(X)
```



Hierarchical Clustering Cont.

1. Parameters:

- **n_clusters=5**: Specifies that the data will be divided into 5 clusters.
- **affinity='euclidean'**: The distance metric used. Here, it's set to 'euclidean', which is appropriate for Ward's method.
- **linkage='ward'**: Specifies Ward's method, which is effective for minimizing variance within clusters.

2. fit_predict(X):

- This method both fits the model to data X and returns an array y_hc, where each element corresponds to the cluster label assigned to each sample in X.



Hierarchical Clustering Cont.

✓ Visualising the clusters

```
[6] plt.scatter(X[y_hc == 0, 0], X[y_hc == 0, 1], s = 100, c = 'red', label = 'Cluster 1')
plt.scatter(X[y_hc == 1, 0], X[y_hc == 1, 1], s = 100, c = 'blue', label = 'Cluster 2')
plt.scatter(X[y_hc == 2, 0], X[y_hc == 2, 1], s = 100, c = 'green', label = 'Cluster 3')
plt.scatter(X[y_hc == 3, 0], X[y_hc == 3, 1], s = 100, c = 'cyan', label = 'Cluster 4')
plt.scatter(X[y_hc == 4, 0], X[y_hc == 4, 1], s = 100, c = 'magenta', label = 'Cluster 5')
plt.title('Clusters of customers')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```

Hierarchical Clustering Cont.

Explanation

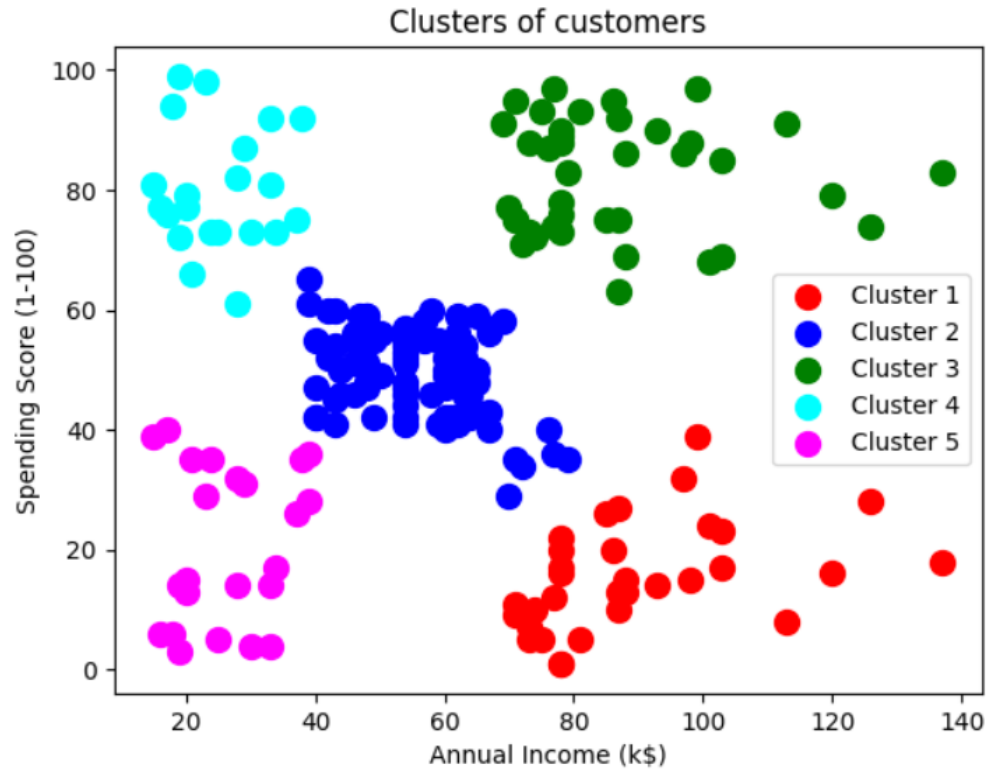
- **`X[y_hc == i, 0]` and `X[y_hc == i, 1]`**: These slices select points in cluster *i*, where `y_hc == i` filters the points in cluster *i*, and `[0]` and `[1]` access specific feature columns (e.g., Annual Income and Spending Score).
- **`s=100`**: Sets the marker size.
- **`c='color'`**: Specifies the color for each cluster.
- **`label='Cluster n'`**: Labels each cluster for the legend, making the plot easier to interpret.

Result

After running this code, you'll get a scatter plot with five distinct clusters, each marked by a different color and labeled accordingly. The `legend()` call will display a legend, making it clear which color corresponds to each cluster.



Hierarchical Clustering Cont.



Hierarchical Clustering Cont.

```
[7] dataset["cluster_hc"]=y_hc
```

dataset

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	cluster_hc
0	1	Male	19	15	39	4
1	2	Male	21	15	81	3
2	3	Female	20	16	6	4
3	4	Female	23	16	77	3
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	2
196	197	Female	45	126	28	0
197	198	Male	32	126	74	2
198	199	Male	32	137	18	0
199	200	Male	30	137	83	2

200 rows × 6 columns



Hierarchical Clustering Cont.

```
[10] from sklearn.metrics import silhouette_score  
     silhouette = silhouette_score(X, y_hc)  
     print(silhouette)
```

⇒ 0.5529945955148897

Silhouette Score:

1. Measures how similar a point is to its own cluster compared to other clusters. Ranges from -1 to 1.



Silhouette Score

Understanding Silhouette Score:

- **Range:** The Silhouette Score ranges from -1 to 1 :
 - **Close to 1:** Points are well clustered; they are far away from neighboring clusters.
 - **Close to 0:** Points are on or very near the decision boundary between two neighboring clusters.
 - **Close to -1 :** Points may have been assigned to the wrong cluster, being closer to points in another cluster.





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Thank you

